

Austin Zhai

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EDUCATION

University of British Columbia

Bachelor of Applied Science in Computer Engineering

- Cumulative Average: 83.7% | GPA: 3.90/4.33 | Honors: Dean's List

Vancouver, BC

Graduation Date: April 2029

EXPERIENCE

IoT Cyber Defense Extern

Remote

Hydroficient

April 2026 – Present

- Engineered and stress-tested a 5-layer IoT security pipeline (TLS, mTLS, HMAC signing, timestamp validation, sequence counters) for a simulated 500-room hotel water management system, reducing unauthorized broker access to zero
- Detected and defeated 3 live attack classes (eavesdropping, sensor spoofing, replay attacks) by building a Python/Mosquitto MQTT pipeline,, then deploying an Isolation Forest anomaly detection model integrated into a real-time Streamlit monitoring dashboard

Bilingual Data Analyst

Remote

TELUS Digital

March 2026 – Present

- Improve accuracy of AI-powered mapping products used by millions globally by evaluating **80+** bilingual map queries weekly across French and English, maintaining a **~90%** accuracy rate across POI validity, search relevance, and location intent assessment
- Accelerate multilingual model retraining by identifying and flagging discrepancies in French and English map data across address components, pin placement, and autocomplete suggestions using standardized evaluation rubrics

Operations Team Member

Vancouver, BC

UBC Sailbot

September 2025 – April 2026

- Increased social media engagement by **50-100%** and contributed to securing **18** active sponsors by independently managing a brand campaign spanning content planning, graphic design, and video production
- Improved cross-subteam coordination across electrical, mechanical, and software divisions by designing and maintaining technical documentation systems for integration workflows, design updates, and testing schedules

Founder and Operator, Independent Haircutting Business

Calgary, AB

Self-employed

August 2024 – December 2025

- Grew a client base generating **\$7,000+** in revenue across **350+** appointments by implementing a word-of-mouth referral incentive system that converted existing clients into active promoters
- Achieved an **~85%** retention rate by designing an Excel-based management system tracking appointments, payments, client history, and follow-up communications across a **50+** person client base

PROJECTS

Portfolio Vision

[Github](#)

React, JavaScript, Python, Supabase (PostgreSQL), Google Cloud, Vercel

May 2026

- Built and deployed a full-stack ETF portfolio decomposer used by **400+** visitors in the first week, by developing a recursive decomposition engine in React covering **500+** tickers with USD/CAD conversion via Frankfurter API
- Reduced API usage by **~95%** per session by implementing a 24-hour localStorage price cache, by integrating Twelve Data's stock price API with per-ticker TTL-based cache invalidation
- Enabled secure multi-user portfolio saving by implementing Google OAuth and OTP email authentication with row-level security policies, by configuring Supabase Auth and a Google Cloud OAuth 2.0 project with authorized domain management
- Shipped a production web application for desktop and mobile devices from zero to custom domain in under 2 weeks, by building with React, Vite, Tailwind CSS, and Vercel with environment-based secrets and auto-deploy on push

Servo Sonar Radar

[Github](#)

Arduino, C++, Processing

May 2026

- Built a real-time object detection system with a 180° sweep range and 3-mode state machine (auto-sweep, pause-on-detect, joystick control) by mounting an HC-SR04 ultrasonic sensor on a servo motor with millis()-based non-blocking timing
- Visualized live sonar data on a custom radar display rendering angle, distance, and proximity alerts in real time by implementing a Serial communication pipeline between Arduino and Processing at 9600 baud

SKILLS & INTERESTS

Languages: Python, C, Arduino, C++, JavaScript, HTML/CSS, Java

Technologies and Frameworks: React.js, Node.js, Vite, Tailwind CSS, Git/GitHub, Supabase (PostgreSQL), REST APIs

Interests: Personal Finance, Weight Training, Basketball/Sports, Piano